

## CURRICULUM VITAE

Dept. of Mathematical Sciences  
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I am a **computational biologist** passionate about developing quantitative techniques to understand the biological principles governing disease in order to improve clinical decision making.

## WORK EXPERIENCE

**Assistant Professor at Chalmers University of Technology (2023–)**      **Gothenburg, Sweden**  
Head of Computing Disease Evolution in cancer (CODEc) group

**Postdoctoral Researcher at Barts Cancer Institute (BCI) and Institute of Cancer Research (ICR) (2017–2022)**      **London, United Kingdom**

## EDUCATION

**Docent in Computational Biology (2025)**

Chalmers University of Technology

Göteborg, Sweden

**Ph.D. in Theoretical Systems Biology (2013–2017)**

Imperial College London (2013–2017)

London, United Kingdom

Supervisor: Prof Michael P.H. Stumpf

**M.Sc. of Infobionics Engineer (2012–2013)**

Pázmány Péter Catholic University

Budapest, Hungary

**B.Sc. in Molecular Bionics (2008–2012)**

Pázmány Péter Catholic University

Budapest, Hungary

## GRANTS &amp; AWARDS

WASP Academic Doctoral Student grant, *co-applicant* (2025–2030)      4,000,000 SEK

MSCA Doctoral Training Network grant, *co-applicant* (2026–2029)      3,250,000 SEK

Swedish Foundations Starting Grant (2025–2031)      13,600,000 SEK

Barncancerfonden project grant (2025–2027)      1,000,000 SEK

Vetenskapsrådet starting grant (2024–2029)      4,400,000 SEK

Area of Advance Health Engineering seed grant (2024)      150,000 SEK

Area of Advance Health Engineering Assistant Professor start-up grant (2022)      10,000,000 SEK

Schrödinger Scholarship (2013–2016)      ~ 1,000,000 SEK

Travel grants from SVEFUM, Adlerbertska, Centre for Computational Biology      ~ 100,000 SEK

Emerging Leaders in Computational Oncology Award for scientific achievement (2020)

## SUPERVISION &amp; TEACHING

**PhD, main supervisor:** Lotta Eriksson (2024–)

**PhD, co-supervisor:** Maiya Tebäck (2025–); Chen Gu (2025–)

**MSc dissertation, main supervisor:**

Pablo Vadillo & Violant Moreno (2025); Anna Fehér Svensson & Samuel Kronquist (2025); Ni Zuoyu (2024); Hampus Johansson (2024); Linnea Hallin & Lotta Eriksson (2024); Charlotte Preunkert & Simone Piccioni (2024); Gerard Llibre (2021); Syeda Fatima (2021); Bryony Coppack (2020); Dr Badia Ahmed (2021); Jay Stanley (2019)\*; Sisi Fan, Quentin Geissmann, Saulius Lukauskas (2015)\*\*

\*Jay received the Best MSc Dissertation prize and presented his project at the British Society of Gastroenterology Annual Meeting (2020).

\*\*The project by Sisi, Quentin & Saul was published in *Bioinformatics* (2016).

### **Independent lecturer, examiner and course responsible:**

Mathematical models for biology (2025); Statistical analysis and experiment design (2023–); Quantifying immune selection from cancer genomics (2022); Stochastic modelling for biology (2022); Genomics for medicine (2022); Coding for Medics (2018–2019)

### **Teaching assistant:**

Problem-based learning for medicine (2019–2020); Coding for Medics (2016–2017) Introduction to modelling and Matlab (2016); Introduction to Bioinformatics module (2013–2015); Genome and protein databases (2013–2015); Digital Signal Processing (2013); Operation of AFM and STM (2013); Data Mining (2012); Physics of Information Technology (2011); Cell Biology and Genetics I-II (2010); Calculus I-II (2009–2011); Linear Algebra and Discrete Mathematics I-II (2009–2012)

## **ACADEMIC CITIZENSHIP & POSITIONS OF TRUST**

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**Associate Editor for npj Precision Oncology (2024–2025)**

**Guest Associate Editor for Frontiers in Genetics (2021–2022)**

Journal Research Topic: Computational approaches applied to cancer genetics, immunogenomics, and immuno-oncology

**Lead of Immune Analysis working group (2018–)**

Within the 100,000 Genomes Project Colorectal, Endometrial and Testicular GeCIPs

**External evaluator for Innovative Medizinische Forschung Münster (2021)**

**Expert reviewer (2016–)** for the following journals:

Nature Communications Biology, npj Precision Oncology, OncoImmunology, Genome Medicine, International Journal of Quantum Chemistry, Frontiers in Genetics, Journal of Pathology, Nature Ecology & Evolution, BMC Bioinformatics

**Opponent for PhD defence (2024)** for Ragnhild Laursen, Aarhus University

**Opponent for midway seminar (2024)** for Malin Ramme, Center for Bionics and Pain Research

## **INVITED & OUTREACH PRESENTATIONS**

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|                                                                         |                              |
|-------------------------------------------------------------------------|------------------------------|
| Somatic Evolution and Tumour Microenvironment (2024)                    | <b>London, UK</b>            |
| Data-driven mechanistic mathematical modelling for life sciences (2023) | <b>Göteborg, Sweden</b>      |
| Modelling for Biology and Medicine (2023)                               | <b>Linköping, Sweden</b>     |
| Nordic Conference in Mathematical Statistics (2023)                     | <b>Göteborg, Sweden</b>      |
| Evolutionary Biology and Ecology of Cancer Summer School (2022)         | <b>Cambridge, UK</b>         |
| Cancer, Multicellularity and Complex Systems (2022)                     | <b>Lund, Sweden</b>          |
| Statistical Physics Approaches to Systems Biology (2019)                | <b>Havana, Cuba</b>          |
| SIAM Conference on Uncertainty Quantification (2016)                    | <b>Lausanne, Switzerland</b> |

**Vetenskapsfestivalen (2024, 2026)** presentation as part of Matematisk afton

**Pink Sunday (2024, 2025)** workshop on programming to women and non-binary persons

**Panel discussion (2023)** on science and art careers at AHA! Festival

**Scientific Society (2022)** (Tudós Társaság) at Embassy of Hungary in London

## **CAREER BREAKS**

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Maternity leave (2025–)

## LIST OF SELECTED PUBLICATIONS

**No. of publications:** 32 [21 peer-reviewed published, 1 peer-reviewed accepted, 1 preprint, 9 proceedings]

**No. of citations:** 1448 (*Google Scholar*)

**h-index:** 15 (*Google Scholar*)

\*: equal contribution; †: corresponding author

1. L. Eriksson and **E. Lakatos**<sup>†</sup>: “Sensitive detection of copy number alterations in low-pass liquid biopsy sequencing data,” *Briefings in Bioinformatics*, 2026
2. **E. Lakatos**<sup>\*,†</sup>, V. Gunasri<sup>\*</sup>, L. Zapata, *et al.*: “Epigenetically driven and early immune evasion in colorectal cancer evolution,” *Nature Genetics*, 2025
3. A. Cornish<sup>\*</sup>, A. Gruber<sup>\*</sup>, B. Kinnersley<sup>\*</sup>, D. Chubb<sup>\*</sup>, A. Frangou<sup>\*</sup>, G. Caravagna<sup>\*</sup>, B. Noyvert<sup>\*</sup>, **E. Lakatos**<sup>\*</sup>, H. Wood<sup>\*</sup>, C. Arnedo-Pac<sup>\*</sup>, *et al.*: “The genomic landscape of 2,023 colorectal cancers,” *Nature*, 2024
4. H. Kayhanian<sup>\*</sup>, W. Cross<sup>\*</sup>, S. E. M. van der Horst<sup>\*</sup>, P. Barmapoutis<sup>\*</sup>, **E. Lakatos**<sup>\*</sup>, *et al.*: “Homopolymer switches mediate adaptive mutability in mismatch repair-deficient colorectal cancer,” *Nature Genetics*, 2024
5. Q. Guo, **E. Lakatos**, I. Al Bakir, K. Curtius, T. A. Graham and V. Mustonen: “The mutational signatures of formalin fixation on the human genome,” *Nature Communications*, 2022
6. **E. Lakatos**<sup>†</sup>, H. Hockings, M. Mossner, W. Huang, M. Lockley and T. A. Graham<sup>†</sup>: “LiquidCNA: tracking subclonal evolution from longitudinal liquid biopsies using somatic copy number alterations,” *iScience*, 2021
7. **E. Lakatos**, M. J. Williams, R. O. Schenck, *et al.*: “Evolutionary dynamics of neoantigens in growing tumors,” *Nature Genetics*, 2020
8. R. O. Schenck, **E. Lakatos**, C. Gatenbee, T. A. Graham and A. R. A. Anderson: “NeoPredPipe: High-Throughput Neoantigen Prediction and Recognition Potential Pipeline,” *BMC Bioinformatics*, 2019
9. **E. Lakatos** and M. P. H. Stumpf: “Control mechanisms for stochastic biochemical systems via computation of reachable set,” *Royal Society Open Science*, 2017
10. **E. Lakatos**, P. Kirk, A. Ale and M. P. H. Stumpf: “Multivariate moment closure techniques for stochastic kinetic models,” *The Journal of Chemical Physics*, 2015